7.2.2 Nested dissection

7.2.2 Nested dissection, Part 1

fit width

The purpose of the game is to limit ***fill-in***, which happens when zeroes turn into nonzeroes. With an example that would result from, for example, Poisson's equation, we will illustrate the basic techniques, which are known as "nested dissection."

If you consider the mesh that results from the discretization of, for example, a square domain, the numbering of the mesh points does not need to be according to the "natural ordering" we chose to use before. As we number the mesh points, we reorder (permute) both the columns of the matrix (which correspond to the elements υ_i to be computed) and the equations that tell one how υ_i is computed from its neighbors. If we choose a ***separator***, the points highlighted in red in Figure 7.2.2.1 (Top-Left), and order the mesh points to its left first, then the ones to its right, and finally the points in the separator, we create a pattern of zeroes, as illustrated in Figure 7.2.2.1 (Top-Right).

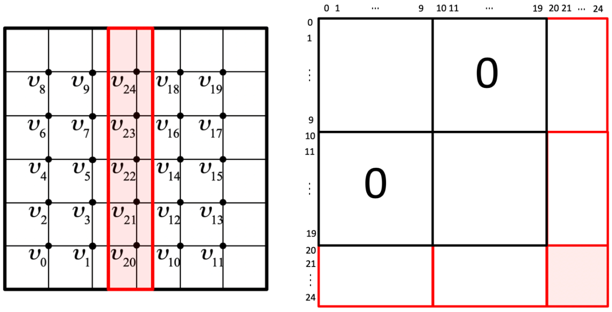

Figure 7.2.2.1. An illustration of nested dissection.

Homework 7.2.2.1. Consider the SPD matrix

$$A = \left(\begin{array}{c|c|c} A_{00} & 0 & A_{20}^T \\ \hline 0 & A_{11} & A_{21}^T \\ \hline A_{20} & A_{21} & A_{22} \end{array} \right).$$

- What special structure does the Cholesky factor of this matrix have?
- How can the different parts of the Cholesky factor be computed in a way that takes advantage of the zero blocks?
- How do you take advantage of the zero pattern when solving with the Cholesky factors?

▾ Solution

- The Cholesky factor of this matrix has the structure

$$L = \left(\begin{array}{c|c|c} L_{00} & 0 & 0 \\ \hline 0 & L_{11} & 0 \\ \hline L_{20} & L_{21} & L_{22} \end{array} \right).$$

- We notice that $A = LL^T$ means that

$$\left(\begin{array}{c|c|c} A_{00} & 0 & A_{20}^T \\ \hline 0 & A_{11} & A_{21}^T \\ \hline A_{20} & A_{21} & A_{22} \end{array} \right) = \underbrace{\left(\begin{array}{c|c|c} L_{00} & 0 & 0 \\ \hline 0 & L_{11} & 0 \\ \hline L_{20} & L_{21} & L_{22} \end{array} \right) \left(\begin{array}{c|c|c} L_{00} & 0 & 0 \\ \hline 0 & L_{11} & 0 \\ \hline L_{20} & L_{21} & L_{22} \end{array} \right)^T}_{\left(\begin{array}{c|c|c} L_{00}L_{00}^T & 0 & \star \\ \hline 0 & L_{11}L_{11}^T & \star \\ \hline L_{20}L_{00}^T & L_{21}L_{11}^T & L_{20}L_{20}^T + L_{21}L_{21}^T + L_{22}L_{22}^T \end{array} \right)},$$

  where the $\star$s indicate "symmetric parts" that don't play a role. We deduce that the following steps will yield the Cholesky factor:

  - Compute the Cholesky factor of A_{00}:

$$A_{00} = L_{00}L_{00}^T,$$

    overwriting A_{00} with the result.
  - Compute the Cholesky factor of A_{11}:

$$A_{11} = L_{11}L_{11}^T,$$

overwriting A_{11} with the result.

- Solve

$$XL_{00}^T = A_{20}$$

for X, overwriting A_{20} with the result. (This is a triangular solve with multiple right-hand sides in disguise.)

- Solve

$$XL_{11}^T = A_{21}$$

for X, overwriting A_{21} with the result. (This is a triangular solve with multiple right-hand sides in disguise.)

- Update the lower triangular part of A_{22} with

$$A_{22} - L_{20}L_{20}^T - L_{21}L_{21}^T.$$

- Compute the Cholesky factor of A_{22}:

$$A_{22} = L_{22}L_{22}^T,$$

overwriting A_{22} with the result.

- If we now want to solve $Ax = y$, we can instead first solve $Lz = y$ and then $L^Tx = z$. Consider

$$\left(\begin{array}{c|c|c} L_{00} & 0 & 0 \\ \hline 0 & L_{11} & 0 \\ \hline L_{20} & L_{21} & L_{22} \end{array} \right) \left(\begin{array}{c} z_0 \\ \hline z_1 \\ \hline z_2 \end{array} \right) = \left(\begin{array}{c} y_0 \\ \hline y_1 \\ \hline y_2 \end{array} \right).$$

This can be solved via the steps

- Solve $L_{00}z_0 = y_0$.
- Solve $L_{11}z_1 = y_1$.
- Solve $L_{22}z_2 = y_2 - L_{20}z_0 - L_{21}z_1$.

Similarly,

$$\left(\begin{array}{c|c|c} L_{00}^T & 0 & L_{20}^T \\ \hline 0 & L_{11}^T & L_{21}^T \\ \hline 0 & 0 & L_{22}^T \end{array} \right) \left(\begin{array}{c} x_0 \\ \hline x_1 \\ \hline x_2 \end{array} \right) = \left(\begin{array}{c} z_0 \\ \hline z_1 \\ \hline z_2 \end{array} \right).$$

can be solved via the steps

- Solve $L_{22}^Tx_2 = z_2$.
- Solve $L_{11}^Tx_1 = z_1 - L_{21}^Tx_2$.
- Solve $L_{00}^Tx_0 = z_0 - L_{20}^Tx_2$.

7.2.2 Nested dissection, Part 2

fit width

Each of the three subdomains that were created in Figure 7.2.2.1 can themselves be reordered by identifying separators. In Figure 7.2.2.2 we illustrate this only for the left and right subdomains. This creates a recursive structure in the matrix. Hence, the name ***nested dissection*** for this approach.

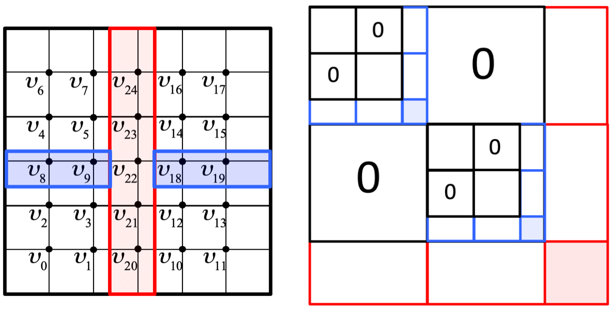

Figure 7.2.2.2. A second level of nested dissection.